

An overview of force measurement: sensing methods, multi-axis measurement, dynamic and static measurement, force estimation and applications.

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Abstract—The abstract goes here.

Index Terms—Interaction Force, Force Measurement.

I. INTRODUCTION

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II. FORCE SENSING METHODES

There are many methods to measure forces exerted on objects. These methods differ from one another in the physical principle of sensing. In this section, we will explain some of these methods and compare between advantages and disadvantages of each method. As shown in Table (??).

A. Resistive force transducer

B. Capacitive force transducer

C. Inductive force transducer

This force transducer is also called search coil sensors, pickup coil sensors, or magnetic loop sensors. This tool converts mechanical force to variation in the inductance. This variation results from a change in impedance, reactance, a change involving a transformer or mutual inductance [1]. The electrical output of inductive force transducer is given, based on Faraday’s fundamental law of induction, by the relationship (1) [2]:

$$E_m = -n \frac{d\Phi}{dt} = -nA \frac{dB}{dt} \quad (1)$$

where E_m is the term for generated or induced voltage, $\phi = BA$ is the magnetic flux passing through a coil with a number of turns n and an area A , B is the external magnetic flux density. The figure(1) shows a type of inductive force sensors. This transformer measures coil impedance (Z) variations during mechanical stressing, being proportional to the applied axial force in the range of $25kN$ [3]. The detect

accuracy of inductive force sensors are in mN level [4]. To increase the sensitivity of an inductive sensor, the coil requires a large effective area and a large number of turns depending on the relationship (1). Consequently, the sensor features a robust, bulky structure, resulting in very low spatial resolution and lower repeatability, particularly when arrayed [2] [7]. Inductive force sensing systems usually have complex mechanical construction and complicated electronic circuit [8]. The inductive F/M sensors always rely on combining a position sensor and a force-to-displacement converter [5] [6]. In the literature, Giesberts et.al designed a small-sized, inductive force sensor for long-term, precise measurements in a portable setting—specifically for medical applications. The first application for treating clubfoot [9]. This sensor measured the force exerted by the foot against the cast, thereby determining the most efficient and effective timing for cast changes. The designed sensor demonstrated high sensitivity ($(1.7 \pm 0.8 \times 10^5)$ counts/N), resolution ((0.15 ± 0.06) mN) and accuracy ($(3.5 \pm 3.0)\%$) for the application. The observed drift was $(2.1 \pm 0.7)\%/\log_{10}(h)$. The second medical application [10] is assistance in the treatment of Dupuytren contracture. Changes in finger contraction forces exerted on the splint were measured over time. A decrease in the measured force was observed, with equilibrium reached after three hours. Measurements taken over several weeks showed that equilibrium was reached earlier.

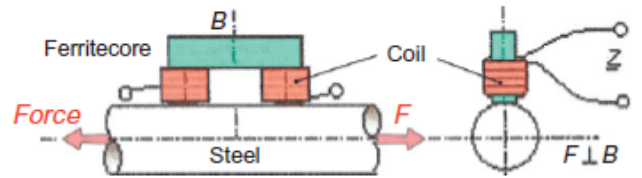


Fig. 1: Force and stress measurement with an eddy-current transducer on a steel bar [11]

D. Piezoelectric force transducer

E. *Fiber optic force transducer*

F. *Vibrating wire force transducer*

III. STATIC VS DYNAMIC FORCE MEASUREMENT

IV. MULTI-DIMENSIONAL FORCE MEASUREMENT

V. FORCE ESTIMATION

VI. FORCE MEASUREMENT APPLICATIONS

A. *Force measurement in monitoring and testing tasks*

B. *Use of force measurement in automated systems*

C. *Force measurement in research and development laboratories*

VII. CONCLUSION

The conclusion goes here.

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